

Future Tech Lab

IINSPARK ROBOTICS – BUILDING TECH-READY MINDS FOR A SMARTER WORLD Welcome to IINSPARK Robotics — a future-driven, tech- integrated learning ecosystem where students don't just learn about machines, they create intelligent systems that solve real-world problems. We're shaping next-gen thinkers, coders, designers, and engineers through a curriculum that combines robotics, artificial intelligence, machine logic, and real-time data processing with structured academic alignment. .; Why Choose IINSPARK Robotics? • Curriculum-aligned & Classroom-complimentary Our robotics modules run parallel to school syllabi and complement concepts in Science, Math, ICT, and Environmental Studies. Students connect classroom learning to practical tech applications. • Real-World Simulations, Real-Time Problem Solving Students work on actual challenges like smart traffic systems, agricultural automation, home automation, disaster response bots, and environment-monitoring devices. • Progressive, Level-Wise Mastery Path Each learner progresses through structured technological complexity, starting from basic logic gates and sensors to full-fledged automation and AI integration. • STEM & NEP Compliant

Designed in line with India's National Education Policy, our program emphasizes experiential, interdisciplinary learning and is mapped to STEM and emerging technology benchmarks. .; ôá Our Robotics Pathway: From Foundations to Innovation`oundation Layer: Human-Machine Interaction • Understanding input-output systems, basic electronic components, and simple automation • Introduction to robotic kits with modular plug-in parts • Logic-building through task-based challenges Outcome: Learners develop a functional understanding of how machines react to stimuli and follow programmed behavior. .; 'V–ÆB Æ yer: Mechatronics & Visual Coding • Constructing robots using gears, motors, sensors, and control units • Programming using block-based languages (Blockly, Scratch, etc.) • Projects include line-followers, obstacle-avoiding bots, and smart lighting systems Outcome: Students transition from theory to creation and understand fundamental engineering principles.

6öFR Æ yer: Embedded Systems & Control Logic • Introduction to microcontrollers (Arduino, Raspberry Pi, ESP32) • Writing code using Python, C++, and real-time input handling • Designing systems for smart agriculture, pollution control, and energy efficiency Outcome: Learners gain deep exposure to core programming and electronics, merging digital logic with hardware. .; –çFVÆÆ–vVæ6R Æ yer: AI, IoT & Machine Learning • Integration of machine learning models, camera vision, object detection • IoT-based projects: remote monitoring, smart homes, voice-activated bots • Data acquisition and decision-making using real-world parameters Outcome: Students begin working with intelligent machines, capable of learning, adapting, and communicating over networks. .; –ææövation Layer: Productization & Entrepreneurship • Capstone projects addressing healthcare, security, sustainability, and mobility

- Building patentable prototypes • Business case development, MVP creation, and industry pitching • Exposure to national and international robotics competitions Outcome: Learners evolve into tech innovators —capable of taking an idea from concept to product, with entrepreneurial confidence. .; Delivery Format • Hybrid Model – Physical lab + app-based simulations • Gamified Learning – Quests, levels, scores, and badges • DIY Robotics Kits – Modular kits for each level with reusable components • Curiosity Labs – Live demonstrations, exhibitions, and competitions • Certification – Skill-based assessment and milestone certification .; 6 Real-World Integration • Urban development: Traffic, waste, energy optimization • Rural upliftment: Soil moisture sensors, irrigation automation • Health sector: Medicine delivery bots, patient-assist systems

- Environmental protection: Air and water quality bots, smart forest monitoring • Defense & Disaster: Drones, search and rescue bots, surveillance systems .; SCAN TO EXPLORE | SCAN TO INNOVATE This isn't just a robotics course. This is where tomorrow's technologies meet today's learners. Explore the IINSPARK Robotics universe. Join the mission to make India robotics-literate, innovation-first, and future-ready. IINSPARK Connect@iinspark.com +91 8484854683